

Math 421

Friday, November 14

Announcements

- Makeup exam this evening –
please avoid discussing exam

Chapter 3: Continuity

Section 18: Properties of Continuous Functions

Bounded Functions

Definition: A real-valued function f is said to be **bounded** if

$\{f(x) : x \in \text{dom}(f)\}$ is bounded.

i.e. there exists $M \in \mathbb{R}$ such that $|f(x)| \leq M$
for all $x \in \text{dom}(f)$.

- ❖ Give an example of a function that is bounded on $\mathbb{R}$ and a function that is unbounded on $\mathbb{R}$.



Discussion Questions

- ❖ What would it mean for a real-valued function to be *unbounded*?

There exists $x \in \text{dom}(f)$ such that $|f(x)| > M$ for all $M \in \mathbb{R}$

✱ an unbounded function can be unbounded above or unbounded below.

- ❖ Give an example of a bounded function.

$$f(x) = \sin(x) \quad f(x) = 1$$

- ❖ Give an example of an unbounded function.

$$f(x) = x \quad f(x) = \frac{1}{x}$$

- ❖ Give an example of a function that is bounded below, but not bounded above. $f(x) = x^2$

Extreme Value Theorem

Theorem 18.1. Let f be a continuous real-valued function on a closed interval $[a, b]$. Then

- f is a bounded function.

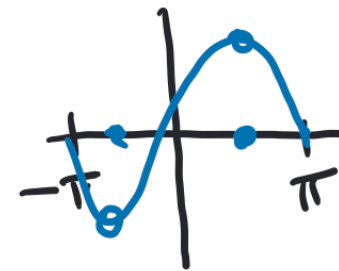
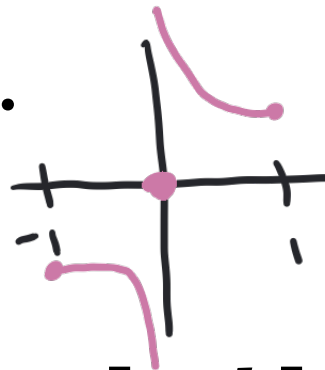
$\left[\begin{array}{l} \bullet f \text{ assumes its maximum and minimum values on } [a, b]. \\ \text{EVT} \end{array} \right.$

i.e. there exist x_0, y_0 in $[a, b]$ such that

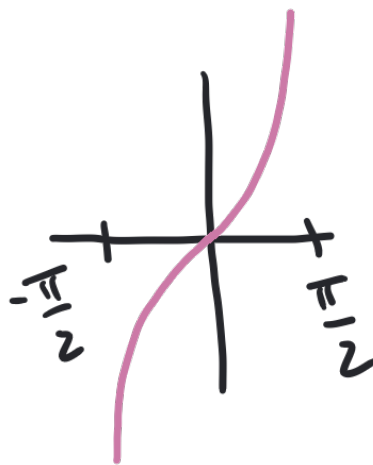
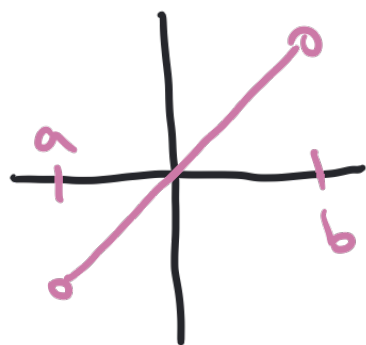
$$\text{minimum value } \boxed{f(x_0)} \leq f(x) \leq \boxed{f(y_0)} \text{ Maximum value for all } x \in [a, b].$$

Discussion Questions

- ❖ Remove the continuity restriction on f from the hypotheses of the theorem. Does the conclusion still hold? If not, find (or sketch) a counterexample.



- ❖ Replace the closed interval, $[a, b]$, with the open interval, (a, b) . Does the conclusion still hold? If not, find (or sketch) a counterexample.

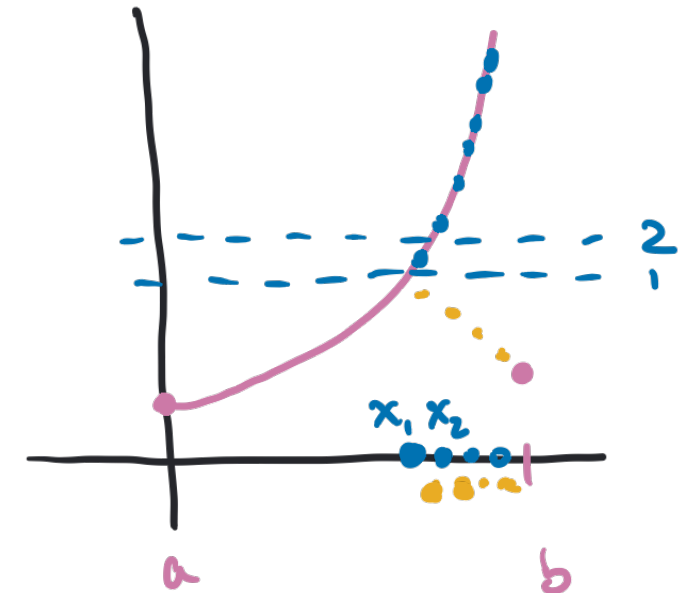


continuous on $[a, b] \Rightarrow$ bounded on $[a, b]$

Proof Idea

$$x_1 \in [a, b], |f(x_1)| > 1$$

$$x_2 \in [a, b], |f(x_2)| > 2$$



Boundedness: for all $M \in \mathbb{R}$, there exists $x \in [a, b]$ s.t. $|f(x)| > M$

Use contradiction. Assume $f: \mathbb{R} \rightarrow \mathbb{R}$

for all $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ continuous s.t.

bounded:
 $x_n \in [a, b]$

on $[a, b]$ and unbounded on $[a, b]$. Construct a $|f(x_n)| > n$

sequence (x_n) , $x_n \in [a, b]$, such that

$\lim |f(x_n)| = \infty$. Find a subsequence (x_{n_k})

with $\lim x_{n_k} = x_0 \in [a, b]$, by continuity $\lim f(x_{n_k}) = f(x_0)$

but $\lim |f(x_{n_k})| = \infty$. Contradiction.

$(f(x_{n_k}))$ is a subsequence $(f(x_n))$

Bolzano-Weierstrass:

bounded sequence has convergent subsequence.